

Homework for Chapter 4

Calculate the spectra of reflection for a single surface for no damping, weak ($\gamma=0.1 \Delta_{LT}/\hbar$) and strong damping ($\gamma=0.6 \Delta_{LT}/\hbar$). Compare the shift of the reflection maxima and minima with respect to the transverse and longitudinal eigenenergies ω_b and ω_l , respectively. Which quantity can be deduced with reasonable accuracy from a first inspection of the reflection spectra?

Below is the optical phonon frequency in GaAs

Phonon Type	Abbreviation	Frequency (THz)	Frequency (cm ⁻¹ , wavenumber)
Transverse Optical	TO	~8.1	~270
Longitudinal Optical	LO	~8.9	~297

The static dielectric constant of GaAs is 12.9

$$\frac{\epsilon_s}{\epsilon_b} = \frac{\omega_L^2}{\omega_0^2} > 1 \quad \text{for } f > 0$$

$$\epsilon_s = \epsilon_b + \frac{f}{\omega_0^2}$$

$$\epsilon(\omega) = \left(\epsilon_b + \frac{f(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} + i \frac{\omega \gamma f}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} \right)$$

$$= \epsilon_1(\omega) + i\epsilon_2(\omega)$$

$$n(\omega) = \left(\frac{1}{2} \left\{ \epsilon_1(\omega) + [\epsilon_1^2(\omega) + \epsilon_2^2(\omega)]^{1/2} \right\} \right)^{1/2}$$

$$\kappa(\omega) = \left(\frac{1}{2} \left\{ -\epsilon_1(\omega) + [\epsilon_1^2(\omega) + \epsilon_2^2(\omega)]^{1/2} \right\} \right)^{1/2}$$

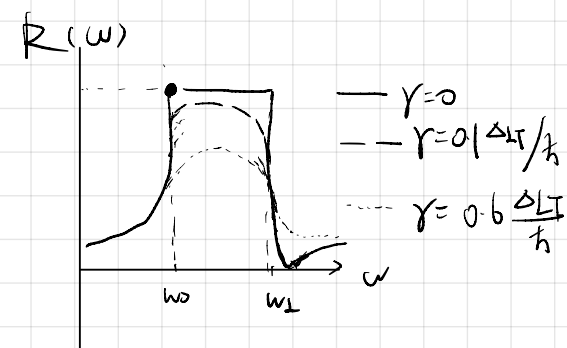
$$R(\omega) = \frac{I_r}{I_i} = \frac{[n(\omega) - 1]^2 + \kappa^2(\omega)}{[n(\omega) + 1]^2 + \kappa^2(\omega)}$$

$$\Delta_{LT} = 8.9 - 8.1 = 0.8 \text{ THz}$$

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$$\zeta_b = \zeta_s \frac{\omega_0^2}{\omega_L^2} = 10.7$$

$$f = (\zeta_s - \zeta_b) \omega_0^2 = 144.3 \text{ THz}^2$$



$R(\omega)$ 如图, 对越大的阻尼, 曲线越平滑

① $\gamma=0$

对 ω_0 可直接得出.

$$\zeta(\omega_0) = \infty$$

$R(\omega)$ 发散, 可以从图中读出突变点

对 ω_L , $\zeta \rightarrow 0$

此时 $R(\omega)$ 出现波谷, 可以大致读出 ω_L , 不能精确

② $\gamma = 0.1 \frac{\Delta_{LT}}{h}$ 弱阻尼

峰值, 波谷的 ω 略微偏大

③ $\gamma = 0.6 \frac{\Delta_{LT}}{h}$

偏移进一步发生, 几乎无法读出数值